Taking Control of the Controls

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Learn Objectives

- will learn about the various selectors available in UiPath to extract and take action on controls:
  - Finding and attach windows
  - Find controls
  - Techniques to wait for a control
  - Acting on controls - mouse and keyboard activities
  - Working with UiExplorer
  - Handling events
  - Screen Scraping
  - When to use OCR
  - Types of OCR available
  - How to use OCR

Finding and attaching windows

- The Attach Window activity can be found in the Activities panel.
- This activity is generally used to attach an already opened window.
  - It is also auto-generated when we record actions using the Basic or Desktop recorder

Implementing the Attach Window activity

- 1. Create a blank project and give it a meaningful name.
- 2. Drag and drop a Flowchart activity on the Designer panel. Also, drag and drop a Click activity inside the Designer panel. Set this Click activity as the Start node.
- 3. Double-click on the Click activity and then click on Indicate on screen. Locate the Notepad icon.
- 4. Drag and drop the Attach Window activity on the main Designer panel. Connect the Attach Window activity to the Click activity.
- 5. Double-click on the Attach Window activity. Click on Click Window on Screen and indicate the Notepad window
- 6. drop the Type into activity, inside the Attach Window activity. Click on the Indicate element inside window and locate the Notepad window where you want to write the text
- 7. Run

Finding the control

- many activities which can be used to find controls on screen/ applications
- the activities that help in finding the controls:
  - Anchor base
  - Element Exists
  - Element scope
  - Find children
  - Find element
  - Find relative element
  - Get ancestor
  - Indicate on screen

Anchor base

- is used for locating the UI element by looking at the UI element next to it.
- This activity is used when we have no control over the selector
- when we do not have a reliable selector, then we should use the Anchor base control to locate the UI

Implement Anchor base

- 1. Drag and drop a Flowchart activity and also drag and drop an Anchor base control from the Activities panel. Connect the Anchor base control with Start
- 2. Double-click on the Anchor base control
- we have to supply to the Anchor base control:
  - Anchor and
  - action activities.
  - Drop find element activity in anchor
  - Drop Type into activity in the action activities
- It will search relative element nearby the element on which want to perform the action

Element Exists

- is used to check the availability of the UI element.
- It checks if the UI element Exists or not.
  - It also returns a Boolean result if the UI Element Exists, then it returns true:
  - otherwise, it returns false
- It is good practice to use this control for UI elements whose availability is not confirmed or those that change frequently
- Use
  - drag and drop the Element Exists control from the Activities panel.
    - Double-click on it. You can see there is an Indicate on screen option.
    - Click on it to indicate the UI element.
  - It returns a Boolean result, which you can retrieve later from the Exists property.
  - You just have to supply a Boolean variable in the Exists property in the Properties panel.

Element scope

- is used to attach a UI element and perform multiple actions on it.
- You can use a bunch of actions within a single UI element
- Drag and drop the Element scope control and double-click on this control
  - you have to indicate the UI element by clicking on "Indicate on screen" and
  - specifying all the actions that you want to perform in the "Do sequence"

Find children

- used to find all the children UI elements of a specified UI element.
- It also retrieves a collection of children UI elements.
- You can use a loop to inspect all the children UI elements or set up some filter criteria to filter out the UI elements
- Use
- Drag and drop the Find children control from the Activities panel
- Double-click on it to indicate the UI element that you want to specify. You can indicate it by clicking on Indicate on screen
- You have to supply a variable of type Ienumerable<UIElements> in the children property

Find Element

- This control is used to find a particular UI element. It waits for that UI element to appear on the screen and returns it back.
- Use
- Just drag and drop this control, and indicate the UI element by clicking on Indicate on screen.
- You can specify the variable of type UI element in property to check the output

Find Relative Element

- This control is similar to the Find element control.
- it uses the relative fixed UI element to recognize the UI element properly.
- This control can be used in scenarios where a reliable selector is not present.
- Just drag and drop this control, and indicate the UI element by clicking on Indicate on screen.
- You can also look for its selector property after indicating the UI element for better analysis

Get ancestor

- This control is used to retrieve the ancestor of the specified UI element.
- You have to supply a variable to receive the ancestor element as output.
- You can specify the variable name in the Ancestor property of the Get ancestor control.
- After receiving the ancestor element, you can retrieve its attributes, properties, and so on for further analysis

Indicate on screen

- This control is used to indicate and select the UI element or region at runtime.
- It gives flexibility to indicate and select the UI element or region while running the workflow
- Do not confuse this with Indicate on screen written inside any activity
- This button is used to locate the region or UI element before the execution of the workflow, while the Indicate on screen control executes its process after the execution of the workflow

Techniques for waiting for a control

- They are:
- 1. Wait Element Vanish
- 2. Wait Image Vanish
- 3. Wait attribute

Wait Element Vanish

- used to wait for a certain element to disappear from the screen
- 1. Create a Blank project and give it a meaningful name.
- 2. Drag and drop a Flowchart activity on the Designer panel. Also, drag and drop the Wait Element Vanish activity on the Designer panel. Set this activity as the Start node.
- 3. Double-click on the Wait Element Vanish activity, then indicate on the screen which element needs to vanish.

Wait Image Vanish

- activity is used to wait for an image to disappear from the UI element
- difference between the Wait Element Vanish and the Wait Image Vanish activities is that the former is used to wait for an element to disappear, while the latter is used to wait for an image to disappear.

Wait attribute

- This activity is used to wait for the value of the specified element attribute to be equal to a string. We have to specify the string explicitly
- 1. Drag and drop a Flowchart activity on the Designer panel. Next, drag and drop the Wait attribute on the Designer panel. Now, right-click on the Wait attribute activity and set it as the Start node.
- 2. Double-click on the Wait attribute activity. We have to specify three values: attribute, element, and text property. We also have to specify the element on which we have to supply the value:
- 3. Run

Act on controls - mouse and keyboard activities

- we have to work with various types of controls, such as Find control, mouse control, keyboard control, and so on, to automate tasks
- Mouse activities
- Those activities that involve interaction with the mouse fall under the category of mouse activities. Which include
  - Click activity
  - Double-click activity
  - Hover activity
- Keyboard activities
- Keyboard activities generally involve an interaction with a keyboard.
- In UiPath Studio, the following are keyboard activities:
  - Send hotkey
  - Type into
  - Type secure text

The Click activity

- When we have to click on a UI element on the screen, we generally use the Click activity.
- Implementation
- 1. Drag and drop a Flowchart on the Designer panel. Search for mouse in the search bar of the Activities panel. Drag and drop the Click activity. Right-click on the Click activity and select Set as Start Node.
- 2. Double-click on the Click activity. Click on Indicate on screen and indicate the UI element you want to click on

The Double-click activity

- Same as the Click activity. The Double Click activity is similar to the Click activity. It just performs the double-click action.
- Using the Double click activity in your project is almost the same as click.
- You have to use the Double click activity instead of the Click activity and indicate the UI element,

The Hover activity

- The Hover activity is used to hover over a UI element. Sometimes, we have to hover over a UI to perform an action
- 1. Drag and drop a Flowchart on the Designer panel. Search for mouse in the search bar of the Activities panel. Drag and drop the Hover activity. Right-click on the Hover activity and select Set as Start Node
- 2. Double-click on the Hover activity. Click on Indicate on screen to indicate the UI element you want to hover on. That's it. We are done.

Send hotkey

- is used to send keystrokes from the keyboard as an input to the screen
- we will use the Send hotkey activity to scroll the Flipkart main page
- 1. Drag and drop a Flowchart on the Designer panel. Search for keyboard in the search bar of the Activities panel. Drag and drop a Send hotkey activity. Rightclick on the Send hotkey activity and select Set as Start Node.
- 2. Double-click on the Send hotkey activity. Click on the Indicate on screen and indicate the required page. You can assign any key by marking the checkboxes. You can also specify the key by selecting a key from the drop-down list

Type into activity

- This activity is used to type the text into the UI element. It also supports special keys
- similar to the Send hotkey activity.
- send the keystrokes along with the special keys. Special keys are optional
- use this activity by simply dragging and dropping the Type into activity, and specifying the keystrokes and the special keys by clicking on the + icon and choosing the key from the drop-down list
- You also have to Indicate on screen the area where you want the text to be typed

Type secure text

- This activity is used to send secure text to the UI element. It sends the string in a secure way
- 1 Drag and drop the Type secure text activity
- 2 Create a variable of type SecureString. Now, double-click on the Type secure text activity and specify the variable's name in the SecureText property of the Type SecureText activity.
- 3 You also have to indicate on the screen by clicking on Indicate on screen

Working with Ui Explorer

- Double-click on the Type into activity.
- Click on the right side of the Selector property, expand Target property to find Selector property.
- A window will pop up. Click on the Open in UiExplorer button
- A window will pop up. You can see the Selector Editor window. Analyze all the text written there.
- You will notice the title: Untitled-Notepad. You just have to edit this title. Just specify test-Notepad between the quotes:

Handling events

- An event occurs when some action is performed. There are different types of events:
  - Element triggering event
  - Image triggering event
  - System triggering event
- Element triggering events
  - This type of event deals with clicking and keypress events.
- Click trigger
  - This event occurs when a specified UI element is clicked.
  - Before using the Click trigger, we have to use the Monitor events activity.
  - Without Monitor events, the Click trigger cannot be used

- Key press trigger
  - This event is similar to the Click trigger.
  - A Key press trigger event occurs when keystrokes have been performed on some particular UI element.
  - It calls the Event handler when it is triggered.
  - While using Key press trigger event you have to specify the key or combination of keys
  - Indicate the UI element on which you want to perform the action

- Image triggering events
  - The Click image trigger is an image tiriggering event.
  - Click image trigger, as the name suggests, is used for when we click an image.
  - You just have to use the Click image trigger event inside the Monitor event and indicate the image.
  - Upon clicking the indicted image in the Click image trigger event, the event handler will be called

- System triggering events
- The following are System triggering events:
  - Hotkey trigger
  - Mouse trigger
  - System trigger
- Hotkey trigger
  - This event is raised when special keys are pressed.
  - As we have already looked at triggering events, you can use the Hotkey trigger event on your own.
  - You have to use this event inside the Monitor event.
  - Specify the special key or combination of keys.
  - Also, provide the event handler that will be called when the event occurs.

- Mouse trigger
- This event is fired when the mouse button is pressed.
- Use this event inside the Monitor event and specify the Mouse button: Either the left mouse button, middle mouse button, or the right mouse button.
- System trigger
- This event is used when you have to use all of the keyboard events, all of the mouse events, or both.
- In the following screenshot, we have dragged and dropped the System trigger event into Monitor events.
- You can specify the trigger input property

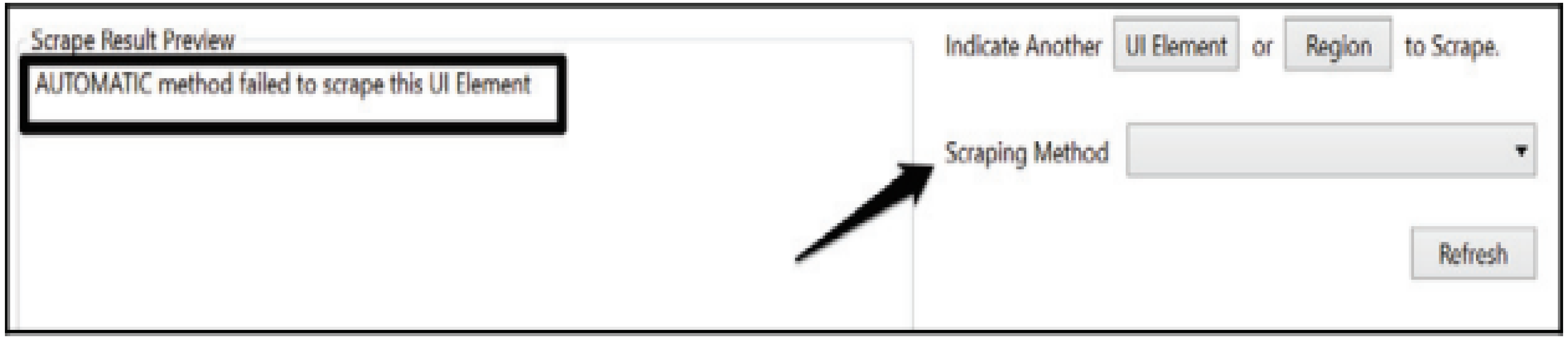
- Screen Scraping
  - Screen Scraping is a method of extracting data from documents, websites, and PDFs.
  - It is a very powerful method for extracting text.
  - We can extract text using the Screen Scraper wizard.
- The Screen Scraper wizard has three scraping methods:
  - Full Text
  - Native
  - OCR

- Full text:
  - The Full text activity is used to extract information from various types of documents and websites.
  - It has a 100% accuracy rate.
  - It is the fastest method among all three methods. It even works in the background.
  - It is also capable of extracting hidden text. However, it is not suitable for Citrix environments.
- Native:
  - This is similar to the Full text method but has some differences. It has a slower speed than the Full text method. It has a 100% accuracy rate, like the Full text method.
  - It does not work in the background.
  - It has an advantage over the Full text method in that it is also capable of extracting the text's position.
  - It cannot extract hidden text. It also does not work with a Citrix environment.

- OCR:
- This method is used when the previous two methods fail to extract information.
- It uses the two OCR engines: Microsoft OCR and Google OCR.
- It has also a scale property: you can choose the scale level as per your need

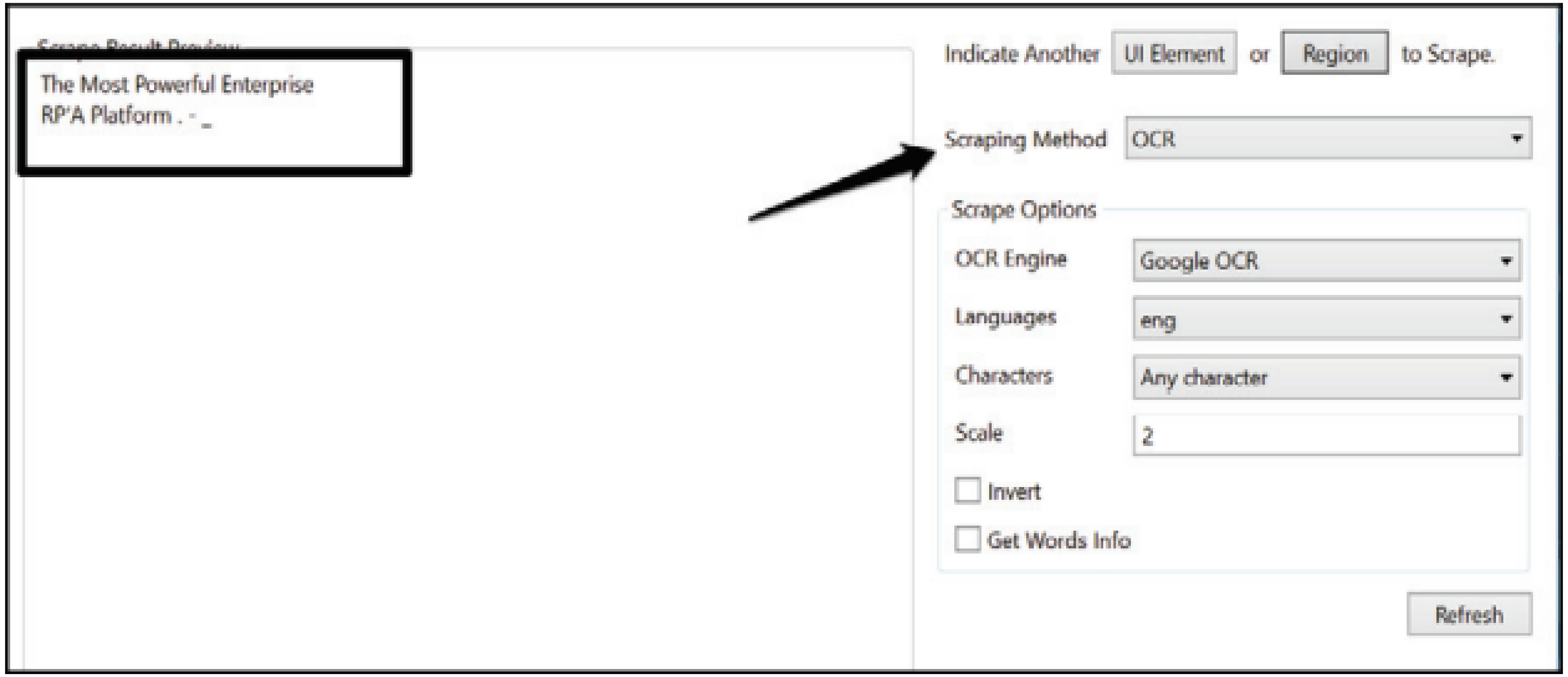
Capability Method	Speed	Accuracy	Background Execution	Extract text position	Extract hidden text	Support for Citrix
Full Text	10/10	100%	Yes	No	Yes	No
Native	8/10	100%	No	Yes	No	No
OCR	3/10	98%	No	Yes	No	Yes

- an example of extracting text from the UiPath website's main page:
- 1. Create a Blank project and give it a meaningful name.
- 2. Log on to the UiPath website by logging in to type url in your browser.
- 3. Drag and drop a Flowchart activity on the Designer panel.
  - Click on the Screen Scraping icon and locate the area from which you want to extract the information. Just choose an area on the UiPath website.
  - A window will pop up stating that the AUTOMATIC method failed to scrape this UI Element.
  - By default, the Screen Scraper Wizard chooses the best scraping method to extract data, but it failed to do so in our case



4. Try choosing another method. We shall choose the Full text method. This too will fail. Next, choose the Native method. This will also fail, as you can see in the following screenshot

5. This time, choose the OCR Scraping method. You can clearly see the extracted text



When to use OCR

- Get Text and Click Text activities fail to extract the text or perform an action.
- This is when OCR comes in, giving us the flexibility to perform actions when existing activities fail to do their job.
- OCR stands for Optical Character Recognition. It is a text recognition technique that transforms printed documents that are scanned into electronic formats.
- OCR is used mainly for images, scanned documents, PDFs, and so on, to extract information or perform actions.
- Extracting information or data from images, scanned documents, or PDFs is a very tedious job.
- Normal activities are not recommended for extracting these types of inputs.

- OCR uses a different method and approach to extract the information.
- There are two OCRs available in UiPath Studio:
- 1. Microsoft OCR
- 2. Google OCR
- Microsoft's OCR is known as MODI, and Google's OCR is called Tesseract.
- OCR is not limited to only these two types of OCR. You are free to use another type of OCR

- 1. Create a Blank project and give it a meaningful name.
- 2. Drag and drop a Flowchart activity on the Designer panel.
  - drag and drop a Get Text activity inside the Designer panel. Now right-click on the Get Text activity and choose Set as Start Node.
- 3. Double-click on the Get Text activity. Click on Indicate on screen.
  - Now indicate the text from which you want to extract information.
  - You have to supply the output value for receiving the text from the Get text activity.
  - Create a GenericValue type of variable and specify the variable name TUS.
- 4. Drag and drop a Message box activity. Connect it to the Get Text activity. Double-click on the Message box activity and specify the variable's name (str) that you created earlier

Types of OCR

- There are two OCRs available in our UiPath Studio: 1. Microsoft OCR 2. Google OCR
- The advantages of Google OCR include the following:
  - Multiple language support can be added in Google OCR
  - Suitable for extracting the text from a small area
  - It has full support for color inversion
  - It can filter only allowed characters
- The advantages of Microsoft OCR include the following:
  - Multiple languages are supported by default
  - It is suitable for extracting text from a large area
  - OCR is not 100% accurate
  - Microsoft and Google's OCRs are not the optimum for every situation

How to use OCR

- 1. Open UiPath Studio and click on a Blank project. Give it a meaningful name. On the Designer panel, drag and drop a Flowchart activity.
- 2. Next, drag and drop a Get OCR Text activity from the Activities panel and set it as the start node.
  - Double-click on it and click on the Indicate on screen option.
  - Choose the specific area from which you want to extract the text from the image.
- 3. Now, click on the Text property of the Get OCR Text activity. Right-click inside the window and choose Create Variable
- 4. Drag and drop the Message box activity. Connect it to the Get OCR with text activity.
  - Double-click on the Message box activity and specify the variable name that you have created earlier in the expression box.